Chapter 4: Random Variables (Sections 4.1 – 4.5)

4.1 Random Variables

When performing an experiment, we are often interested in a specific numerical quantity determined by the outcome, rather than the outcome itself (e.g., the *sum* of two dice rather than the specific pair of numbers).

Definition

A **random variable** is a real-valued function defined on the sample space.

- It assigns a numerical value to each possible outcome of the experiment.
- **Notation:** Usually denoted by capital letters (e.g., X, Y, Z).

Examples

1. **Tossing 3 Coins:**
   - Let Y = number of heads.
   - Outcomes map to values: $(T,T,T) \to 0$, $(H,H,H) \to 3$, etc.
   - Probabilities: $P(Y=0) = 1/8$, $P(Y=1) = 3/8$, etc.
   - Sum of probabilities must equal 1: $\sum P(Y=i) = 1$.
2. **Selecting Balls:**
   - Selecting 3 balls from 20 (numbered 1-20).
   - X = largest number selected.
   - $P(X=i) = \frac{\binom{i-1}{2}}{\binom{20}{3}}$ (Choose ball i, then choose 2 from the $i-1$ smaller numbers).
3. **Independent Coin Flips:**
   - Flip coin (prob p) until Head appears or n flips made.
   - X = number of flips.
   - $P(X=i) = (1-p)^{i-1}p$ for $i < n$.
   - $P(X=n) = (1-p)^{n-1}$ (includes outcomes where nth is H or T).

4.2 Discrete Random Variables

A random variable is **discrete** if it can take on only a countable number of possible values.

Probability Mass Function (PMF)

Defined as $p(a) = P(X=a)$.

- Properties:
  i. $p(x_i) > 0$ for possible values x_i.
  ii. $p(x) = 0$ for other values.
  iii. $\sum_{i=1}^{\infty} p(x_i) = 1$.

Cumulative Distribution Function (CDF)

The function $F(a)$ expresses the probability that X is less than or equal to a.

$$F(a) = P(X \le a) = \sum_{x \le a} p(x)$$

- For a discrete random variable, $F(a)$ is a **step function**. It is constant between possible values and "jumps" by the size of the probability $p(x)$ at each value x.

4.3 Expected Value

The expected value (or mean) is one of the most important concepts in probability, serving as a weighted average of the possible values.

Definition

$$E[X] = \sum_{x} xp(x)$$

- **Interpretation:**
    i. **Weighted Average:** Values with higher probability contribute more to the sum.
    ii. **Frequency Interpretation:** If an experiment is repeated infinitely, the long-run average of the outcomes converges to $E[X]$.
    iii. **Center of Gravity:** Physically, $E[X]$ represents the point where a weightless rod would balance if weights of mass $p(x)$ were placed at points x.

Examples

- **Fair Die:** $E[X] = 1(\frac{1}{6}) + \cdots + 6(\frac{1}{6}) = 3.5$.
- **Indicator Variable:** If $I = 1$ when event A occurs and 0 otherwise:

$$E[I] = 1 \cdot P(A) + 0 \cdot P(A^c) = P(A)$$

- **Quiz Show Strategy:** Comparing expected winnings helps determine optimal decisions (e.g., answering the easier or harder question first based on success probabilities and prize values).

4.4 Expectation of a Function of a Random Variable

If we have a random variable X and want to find the expected value of a function $g(X)$, we do not need to find the PMF of $g(X)$ first.

Proposition 4.1

$$E[g(X)] = \sum_{x} g(x)p(x)$$

- *Meaning:* It is a weighted average of the values $g(x)$, weighted by the probability that $X = x$.

Corollary 4.1 (Linearity)

For constants a and b:

$$E[aX + b] = aE[X] + b$$

- This implies the expectation operator is linear.

Moments

- $E[X]$ is the **first moment** (mean).
- $E[X^n]$ is the n**th moment**.

Applications

- **Stocking Strategy:** Determine the number of units s to stock to maximize expected profit $E[P(s)]$, balancing profit per sale against loss per unsold unit.
- **Utility Theory:** In decision making, one should choose actions that maximize the expected *utility* ($E[u(X)]$) rather than just raw consequences, allowing for weighing risk and preference.

4.5 Variance

While expectation gives the "center" of a distribution, variance measures the "spread" or variation of the values around the mean.

Definition

If X has mean $\mu = E[X]$, the variance is:

$$\mathrm{Var}(X) = E[(X-\mu)^2]$$

Computational Formula

A more practical formula for calculation is:

$$\mathrm{Var}(X) = E[X^2] - (E[X])^2$$

- *Words:* The expected value of the square minus the square of the expected value.

Standard Deviation

$$\mathrm{SD}(X) = \sqrt{\mathrm{Var}(X)}$$

Properties

- **Linear Transformation:** Adding a constant b shifts the distribution but does not change the spread. Multiplying by a scales the spread by a^2 in variance terms.

$$\mathrm{Var}(aX+b) = a^2\mathrm{Var}(X)$$

Example (Fair Die)

- $E[X] = 3.5$
- $E[X^2] = 1^2(\frac{1}{6}) + \cdots + 6^2(\frac{1}{6}) = \frac{91}{6}$
- $\mathrm{Var}(X) = \frac{91}{6} - (3.5)^2 = \frac{35}{12} \approx 2.917$